

ANDY SUKOWSKI-BANG

Mathematics and Computer Science

Kiel, 19/07/2026

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RESEARCH

[1] *Colimits of Enriched Categories*, A. Sukowski-Bang, 2026, DOI. [10.48550/arXiv.2607.07780](https://doi.org/10.48550/arXiv.2607.07780).

Abstract. We introduce categories of strings to give an explicit hom-object formula for colimits of enriched categories. As corollaries, we recover the necklace construction by Dugger and Spivak and describe colimits in $\mathbf{Cat}$ and of Lawvere metric spaces.

EDUCATION

Christian-Albrechts-Universität zu Kiel

Kiel 📍

B.Sc. & M.Sc. Mathematics, minor in Computer Science

since 10/2023

- Bachelor's thesis *Colimits of Enriched Categories* [1], supervised by [Dr. Timo Essig](#), second examiner [Prof. Dr. Richard Weidmann](#)
- Coursework in Higher Homotopy and Category Theory, Algebraic Topology, Differential Geometry, Nielsen-Thurston classification, Geometric Group Theory, Commutative Algebra and Foundations of Algebraic Geometry

University of Rostock (Junior Studies)

Rostock 📍

Imperative Programming in C (2 semesters, passed exam)

2021–2022

William-Stern-Gesellschaft [🔗](#)

Hamburg 📍

Mathematics Program for Gifted Students

2017–2021

Johann-Rist-Gymnasium

Wedel 📍

Physics profile, additional subject Computer Science

2015–06/2023

- Abitur grade 1.1 [📄](#) with awards in Mathematics ([DMV](#)) [📄](#), Physics ([DPG](#)) [📄](#), Comp. Sci. ([GI](#)) [📄](#)
- Taught Computer Science during teacher's parental leave (6 months)
- Led programming club – C, Haskell, Julia, JavaScript

WORK EXPERIENCE

Thales Germany

Kiel 📍

Software Development

11/2023–present

- Multi-agent reinforcement learning with [PPO](#) and [relation-aware Transformer](#) policy
- Partial Ansible to [Rundek](#) transpiler, written in Haskell with [Parsec](#)

Private Tutoring (5 students)

Wedel 📍

Including Mathematics, Computer Science, and Physics

2019–10/2023

Grassau GmbH

Wedel 📍

Linux System Administration and Web Development

08/2021–11/2021

SUMMER SCHOOLS AND CONFERENCES

Stratified Spaces from Higher Category Theory to Applied Topology ↗	Kiel 📍
Registered participant	31/08/2026
Higher Categories for the Working Mathematician ↗	Göttingen 📍
Accepted participant; funding awarded by the Georg-August-University Göttingen	27/07/2026
Young Topologists Meeting 2026 ↗	Copenhagen 📍
Participant; funding awarded by Kiel University	29/06/2026
Higher Structures: Recent Developments and Applications ↗	Hamburg 📍
Participant; funding awarded by University of Hamburg	01/09/2025

SCHOLARSHIPS & COMPETITIONS

- Scholar, [Studienstiftung des deutschen Volkes](#), since 10/2025 [📄](#)
- Scholar, [e-fellows.net](#), since 2023 [📄](#)
- 2nd prize in state round of 59th Mathematics Olympiad, 2020
- Participation in [federal round](#) after 2nd prize in state round of 58th Mathematics Olympiad, 2019

SELECTED PROGRAMMING LANGUAGES & TOOLS

Haskell ([Parsec](#)), C (libpng, ncurses, `<sys/socket.h>`), Julia (Flux.jl, Lux.jl), Python (JAX, Flax, NNX, TensorFlow, PyTorch, Stable Baselines 3), [Typst](#), $\text{\LaTeX}$, Git, GnuPG, Neovim, Linux (Gentoo, Void, Alpine, Arch)

HOBBIES

Programming since 2014 (see [git.andy.sb](#)), climbing/bouldering (7b), electric guitar since 2012 (e.g. in Jazz, Rock, Pop band), ballroom dancing.